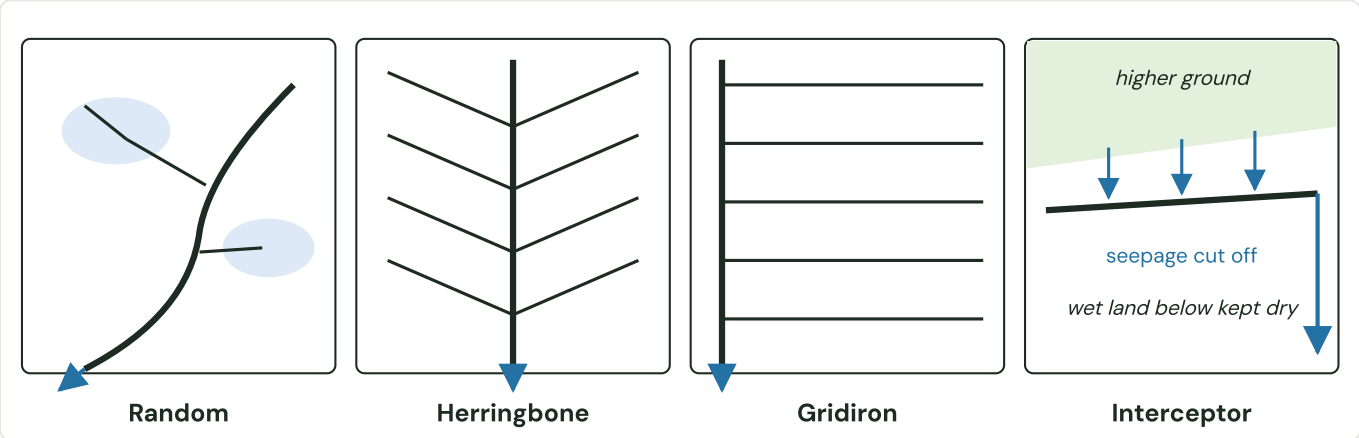
AGRICULTURAL DRAINAGE · NOTE DR06

Tile Drain Layout and Installation — Layouts, Filters, Field Survey, Laying and Supplemental Structures

Course topics: IDE-502 T8, IDE-502 T10

Layout patterns

Tile systems are laid out as random, herringbone, gridiron (parallel) or interceptor (cut-off) systems, often in combination. The first three are relief drainage.



The four basic tile layouts (heavy line = main; light lines = laterals; blue arrow = outlet).

Layout	Use
Random	Scattered wet spots. The main follows the largest natural depression; submains and laterals reach the individual wet areas. Large wet areas may be drained internally on a gridiron or herringbone plan
Herringbone	Parallel laterals enter the main at an angle from **both** sides. Used where the main or submain lies in a narrow depression; drains doubly at the junctions, so it can be less economical. Suits long laterals and thorough drainage
Gridiron (parallel)	Like herringbone but laterals enter from **one** side only. For flat, regularly shaped fields with uniform soil; cheaper than herringbone (fewer junctions and double-drained strips). A **double-main** system in broad flat depressions puts mains on both sides of the waterway to intercept seepage and give laterals an outlet
Interceptor (cut-off)	Intercepts hillside seepage — wetness from horizontal seepage through a permeable layer over an impermeable one. Lay the drain along the bottom of the permeable layer

	Interceptor drainage	Relief drainage
Purpose	Intercepts water coming from outside and diverts it	Removes excess water already in the waterlogged land
Alignment	Perpendicular to groundwater flow; lowers the water table downslope but has little effect upslope	Parallel to groundwater flow, equal drawdown on both sides; series patterns such as gridiron and herringbone
When	Large inflow from a canal, reservoir or higher irrigated land, and the drain can sit in the upper wet area; covered drains preferred; place it upstream of an impermeable barrier	Groundwater inflow is small and interception would not work
Design	As subsurface drainage	Subsurface: as subsurface drainage. Surface: hydrologic design from runoff, hydraulic design by Manning's formula

Filters (envelopes)

- Essential in subsurface drainage — especially in irrigated regions and in non-cohesive soils with < 40% clay — and recommended even in cohesive soils, because faults are hard to correct once buried.
- **Main job:** keep soil particles out of the pipe. Also give the pipe a stable bed and a zone of higher conductivity around it.
- **Mineral filters:** sand, sand–gravel mixtures, and well-graded gravel (most effective). **Organic filters:** crop residues — paddy straw, coir fibre, corn cobs, saw dust.
- **In humid areas** topsoil or permeable trench material may serve as the filter. Joints may be covered with tar-impregnated roofing paper, glass-fibre filter, plastic guards or burlap, placed around the **upper two-thirds** of the tile to stop sediment.
- **A good filter** is more pervious than the base soil (no pressure build-up), with voids small enough to stop soil particles — too-large voids risk clogging or settlement. Minimum thickness 7.5 cm around the tile.

1. Find the gradation curve of the base soil by mechanical analysis.
2. Apply the design limits of the US Bureau of Reclamation and the US Army Corps of Engineers.
3. Choose a filter gradation whose curve is nearly parallel to the base-soil curve.

Stability criterion: (15% size of filter) / (85% size of base soil) < 5, i.e. $D_{15}(\text{filter})/D_{85}(\text{base}) < 5$

Standard companion check (USBR/Terzaghi): for permeability the filter should also satisfy $D_{15}(\text{filter})/D_{15}(\text{base}) \geq 5$.

Field survey

1. Inspect the field and the factors affecting drainage: confirm the soil responds to tile drainage, has no physical limitation, and has a suitable outlet (open ditch or stream).
2. Prepare an accurate contour map (unless inspection fixes the layout). Decide the alignment, type of system, and the location and minimum number of outlets; give tile lines the maximum possible slope. Economy comes from the longest possible laterals and the least main per unit area.

3. Lay laterals along the greatest slope (maximum grade) or across it (to intercept the most groundwater) — topography and soil decide. Plot several layouts and choose the most efficient and economical.
4. **Staking:** stake the mapped lines in the field at 15 m or 7.5 m intervals; number or letter the lines for profiles and the bill of materials.
5. **Grade:** run profile levels along the staked lines, plot tile depth and grade, find the cut at each station from the vertical difference and mark it on the stakes.

- **Line-and-rod method:** stretch a cord at a uniform height above the trench bottom on cross-bars fixed to uprights about 2 m above the trench bottom; measure down with a rod — the grade is right when the vertical gauge just touches the cord.
- **Laser grade control** (large, precise jobs): a thin beam from a rotating prism hits a photo-sensitive receiver on the laying machine, and an electronic circuit adjusts the trencher — a smooth, uniform slope to the outlet with no depressions or sediment traps.

Installation

- **Operations:** set alignment and levels; excavate the trench to depth and slope; place the pipes (with or without envelope); backfill — in the right sequence, with accurate layout transfer, suitable machines and proper grade control.
- **Direction:** start at the downstream outlet and work upstream, so water does not collect in the trench and soften its bottom. Lay drains in straight lines and smooth curves.
- **Hand tools:** tile spade (solid or open, blade 40–55 cm; open for sticky soils), shovel, drain scoop (semi-cylindrical, fits 10–20 cm tiles) and tile hook (12.5 mm round iron).
- **Small-tile trench:** 30–40 cm wide with nearly vertical sides. Dig to 2.5–5 cm above grade and finish the bottom with the scoop so ¼ of the tile circumference rests on firm soil.
- **Trenchless machines:** the vertical plough works like a subsoiler — lifts the soil, forms cracks and installs larger pipes; the V-plough lifts a triangular wedge and installs 10–12.5 cm pipes. Special machines lay pipe lengths up to 300 m at depths up to 3 m.
- **Trenchers:** wheel trenchers (shallow drains; buckets dig and a conveyor throws the soil aside) and chain trenchers (a continuous chain with knives — high capacity, heavy duty, adjustable depth and width, needing careful maintenance).

Depth, cover, spacing and grade

- Depth is measured from the ground to the tile bottom and depends on soil permeability, outlet depth, lateral spacing and depth of the impervious layer. Ideally it keeps the effective water table below the root zone; soil above the water table stays wetter than field capacity because of capillary rise.
- Keep at least 60 cm of soil cover over the tile to resist surface loads and prevent shifting. Sandy soils → deeper drains, wider spacing. Cracking clays take water in fast. Tight clays move water slowly and need shallower, closer drains.

Soil	Spacing	Depth
Clay / clay loam	12–21 m	0.75–0.90 m
Silt loam	18–30 m	1.0–1.2 m
Sandy loam	30–90 m	1.0–1.4 m

- Spacing under 15 m is usually uneconomical except for high-value crops. (This table and Table 11.1 in the design note come from different guidelines; both are kept as printed.)
- **Grade:** use the maximum obtainable slope — laterals along the greatest field slope; mains may be flatter. Minimum grades: 0.10% for 10 cm tile, 0.07% for 12.5–17.5 cm tiles. Maximum ≤ 2% for farm drain tile; on steeper slopes use bell-type sealed sewer pipe. A reduced grade can cause blowouts from hydrostatic pressure and needs a pressure-relief well.
- **Drainage coefficient** (design rate, cm of water per 24 h) depends on soil, crop and how surface water is disposed of; with surface inlets, include the runoff entering the tile. The drainage area usually comes from drain length × spacing, but with surface inlets the coefficient applies to the whole contributing watershed.

Laying, blinding and backfilling

1. Start at the lower end and move upgrade; place tiles ≤ 15 cm in diameter with a tile hook. Lay and blind them as soon as the trench is finished; close the upper end with a brick or stone to keep dirt out.
2. Use manufactured junctions for smooth, tight joints (otherwise chip the tile and seal with mortar). Connect laterals at the mid-point of the main tile; change direction with smooth curves, not sharp bends; cover the outer side of joints with broken tile pieces.
3. **Blinding:** cover with 15–30 cm of loose earth, preferably topsoil (more permeable). In tight soils, add a gravel layer first so the joints do not seal; hay, straw or corncobs are temporary substitutes with a limited life.
4. **Backfilling:** soon after blinding — by hand, buck scraper or bulldozer, whichever is cheapest.

Supplemental structures

For proper working and long life a tile system needs outlet structures, surface inlets, sedimentation basins, junction boxes, breather wells and relief wells.

Structure	Details
Outlet protection	Protect the outlet into an open ditch from erosion and undermining: a stone headwall (sometimes enough), a concrete or masonry headwall with wing walls, or a notched concrete headwall where a grassed waterway and tile discharge together. Outlet pipe of cast iron or corrugated metal ≥ 2.5 m long, preferably the tile's diameter, extending into the stream for firm anchorage. Flat stones or rock against undercutting; iron rods or metal gates keep out rodents and small animals that nest and block the pipe
Flap gate	A hinged gate that closes automatically when the ditch rises above the tile line — floodwater is laden with sediment, which would settle in a stagnant tile — and opens when the ditch falls
Surface inlet	In tight soils water collects in low spots; an inlet lets it into the tile faster than percolation. Prefer subsurface drainage where possible; if used, locate and build it properly and keep grass around it. **Open inlet** (small flows): replace a tile section with sewer pipe, seal joints with mortar and fit a dome-type (beehive) grating against trash and weeds
Masonry surface inlet	For larger flows: a silt trap formed by extending the structure ≥ 45 cm below the tile bottom; large enough to clean easily and cleaned regularly; a coarse, strong screen on top stops sticks, stalks and straw
Blind inlet (French drain)	For small surface flows or loads too silty for an open inlet: backfill a short tile section with stone or gravel, coarsest over the tile and finer towards the surface, at the lowest part of the depression. Clogs with sediment and organic matter in a few years
Sedimentation basin	Traps sediment before it enters the tile — needed where the tile slope must be reduced, surface water enters, or several laterals join the main at one point. Clean before sediment reaches the outlet tile
Junction box (inspection chamber)	At lateral–collector intersections, where several collectors join, or where laterals enter at different levels. Usually a standard pipe chamber ~1 m in diameter for inspection and cleaning; bottom sealed; top ≥ 30 cm below ground in cultivated fields (can be capped and covered); include a sedimentation basin
Breather / relief well	A small vertical riser ≈ 30 cm above ground, about every 400 m on long mains, preferably along fences or field borders. It relieves pressure: if water pressure in the tile exceeds that in the soil, water escapes through the joints, washes soil and displaces tiles. Needed where the main's grade flattens (by more than 1 %) so the flatter section cannot carry the flow, or where added laterals overload the main

Roots: water-loving trees block tile drains; remove them for about 15 m on either side of the line, or else build a closed line with sealed joints.

Sources: ICAR eCourse 'Drainage Engineering' — Dr M. K. Jha (IIT Kharagpur), reviewed by Prof. K. Yellareddy (Director A&R, WALAMTARI, Hyderabad); AgriMoon e-book. US Bureau of Reclamation, Drainage Manual (filter criteria). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Laterals entering the main from both sides at an angle form a

(a) gridiron system
(b) herringbone system
(c) random system
(d) interceptor system

Q2. For flat, regular fields with uniform soil the most economical layout is

(a) random
(b) herringbone
(c) gridiron
(d) interceptor

Q3. Filter stability criterion for tile drains is

(a) D_{15} filter/D_{85} base < 5
(b) D_{85} filter/D_{15} base < 5
(c) D_{50} filter/D_{50} base > 40
(d) D_{10} filter = D_{10} base

Q4. Minimum filter thickness around a tile is

(a) 2.5 cm
(b) 5 cm
(c) 7.5 cm
(d) 15 cm

Q5. Tile laying should start at

(a) the upper end
(b) the middle
(c) the outlet (downstream) end
(d) any convenient point

Q6. Minimum soil cover over a tile drain is

(a) 30 cm
(b) 45 cm
(c) 60 cm
(d) 100 cm

Q7. Breathers/relief wells on long mains are provided about every

(a) 50 m
(b) 150 m
(c) 400 m
(d) 1000 m

Q8. Trees near tile lines should be removed for about

(a) 3 m
(b) 15 m
(c) 50 m
(d) 100 m

Q9. A masonry surface inlet forms a silt trap by extending at least

(a) 15 cm
(b) 30 cm
(c) 45 cm
(d) 90 cm

Q10. Maximum slope for farm drain tile is about

(a) 0.5%
(b) 2%
(c) 5%
(d) 10%

Answers

Q1 — (b) herringbone system
Q2 — (c) gridiron
Q3 — (a) D_{15} filter/D_{85} base < 5
Q4 — (c) 7.5 cm
Q5 — (c) the outlet (downstream) end
Q6 — (c) 60 cm
Q7 — (c) 400 m
Q8 — (b) 15 m — *either side of the line.*
Q9 — (c) 45 cm — *below the tile bottom.*
Q10 — (b) 2% — *Steeper → bell-type sealed sewer pipe.*